

Placement Intelligence Assistant Dataset

Enhanced Multimodal RAG Exercise

SVECW · Department of Information Technology · RAG-ATHON 24

This dataset is designed for students building **Multimodal Retrieval-Augmented Generation (RAG)** systems. It contains **text narratives, structured tables, bar chart images, multi-hop reasoning challenges, conflicting data, temporal queries, and adversarial test questions** — all within a real-world placement intelligence context. Students must handle each content type with appropriate ingestion, chunking, embedding, and retrieval strategies.

Content Modality Map

Section	Content Type	RAG Challenge	Difficulty
Company Profiles (19 cos)	Structured Tables	Table extraction from PDF	Medium
Interview Experiences	Repetitive Free Text	Deduplication, Chunking	Medium
Hiring Distribution Charts	Bar Chart Images	Multimodal / Vision RAG	Hard
Overall Statistics	Multi-row Table	Cross-document aggregation	Hard
Eligibility Filter Section	Conditional Text	Structured filtering queries	Medium
Multi-hop Reasoning Section	Cross-company Text	Multi-document synthesis	Hard
Temporal Trend Section	Time-series Text	Temporal reasoning queries	Hard
Conflicting Data Section	Inconsistent Records	Hallucination detection	Expert
Adversarial Queries Section	Out-of-corpus Questions	Graceful fallback handling	Expert

Section 1: Company Eligibility Profiles

The table below contains placement eligibility criteria for all 19 companies. This is the primary structured knowledge source. A RAG system must correctly extract and compare these values across companies to answer eligibility queries.

Company	Min CGPA	Max Backlogs	Package (LPA)	Bond (Yrs)	Key Topics	Tech Focus
TCS	7.5	0	4.1	0	DSA, System Design	System Design
Infosys	8.0	0	42.9	0	DSA, OOPs	Java
Deloitte	7.7	1	9.6	1	DSA, Aptitude	System Design
Accenture	8.2	0	17.3	2	DSA, Cloud	System Design
Amazon	6.4	1	28.6	2	DSA, C++, LLD	C++
Flipkart	7.8	2	25.3	2	DSA, Python	Python
Google	7.4	0	42.0	1	DSA, Algorithms	Python
Microsoft	6.1	1	21.4	0	DSA, OS, DBMS	C++
Wipro	6.7	1	26.1	1	DSA, System Design	System Design
Cognizant	8.4	0	42.3	2	DSA, Java	Java
Capgemini	7.1	0	38.3	2	DSA, C++	C++
IBM	7.5	2	27.5	0	DSA, Cloud	C++
Adobe	7.5	0	18.3	1	DSA, System Design	System Design
Oracle	7.7	0	17.3	2	DSA, DBMS	Python
SAP	8.4	0	20.7	2	DSA, C++	C++
HCL	8.4	1	28.1	2	DSA, Cloud	Cloud
Tech Mahindra	8.1	2	35.9	1	DSA, System Design	System Design
Qualcomm	7.2	2	41.3	1	DSA, Cloud	Cloud
Intel	7.0	0	41.4	0	DSA, Python	Python
Samsung R&D;	6.3	2	7.6	2	DSA, Java	Java

■ RAG Challenge — Table Extraction

Standard PyPDFLoader often extracts table data as garbled text, merging columns incorrectly. Students should use pdfplumber or camelot for accurate table extraction, then convert rows to structured JSON or natural language passages before embedding.

Section 2: Interview Experience Summaries

■ RAG Challenge — Duplicate Content

The original dataset contains each interview experience repeated 3× per company. This is intentional — students must detect and deduplicate content during ingestion. Embedding near-identical chunks wastes vector store space and distorts retrieval scores.

■ TCS | Technical Focus: System Design

Round	Details
Round 1	Online Assessment: Aptitude + Logical Reasoning + DSA coding (HackerRank, 90 min)
Round 2	Technical Interview: System Design (LLD/HLD), previous projects, data structures
Round 3	Managerial/HR: Behavioural questions, teamwork scenarios, Why TCS?

■ **Tip:** Strong DSA fundamentals, design patterns, OOPs principles. Know your projects end-to-end.

■ Amazon | Technical Focus: C++ / LLD

Round	Details
Round 1	Online Assessment: 2 DSA problems (medium-hard), 45 min
Round 2	Technical Interview 1: Array/DP/Graph problems + time-space complexity analysis
Round 3	Technical Interview 2: LLD (Low Level Design) — design a parking lot system
Round 4	Bar Raiser + HR: Leadership Principles (STAR format answers essential)

■ **Tip:** Amazon's Leadership Principles are non-negotiable. Prepare STAR stories for all 16 principles. DSA difficulty is higher than service companies.

■ Google | Technical Focus: Python / Algorithms

Round	Details
Round 1	Online Assessment: 2 algorithmic problems (competitive programming level)
Round 2	Phone Screen: 1 coding problem + discussion on approach and complexity
Round 3	Onsite Round 1: Graphs + Dynamic Programming
Round 4	Onsite Round 2: System Design (scalable distributed system)
Round 5	Onsite Round 3: Behavioural (Googleyness)

■ **Tip:** Google emphasises algorithmic thinking over implementation speed. Practice on LeetCode Hard. System Design must cover scalability at Google-scale.

■ Infosys | Technical Focus: Java / OOPs

Round	Details
Round 1	Online Assessment: Aptitude + Logical + Verbal + Java MCQ
Round 2	Technical Interview: Core Java, OOPS principles, collections, multithreading
Round 3	HR Round: Career goals, relocation flexibility, Why Infosys?

■ **Tip:** Focus on Java fundamentals, OOPs, and be clear about your career plan. The interview is relatively straightforward for a 42.9 LPA package.

■ Microsoft | Technical Focus: C++ / OS / DBMS

Round	Details
Round 1	Online Assessment: 3 DSA problems (medium difficulty)
Round 2	Technical Interview 1: Data Structures (Trees, Graphs), problem-solving approach
Round 3	Technical Interview 2: OS concepts (threading, deadlocks), DBMS (indexing, normalization)
Round 4	HR Round: Culture fit, behavioural questions

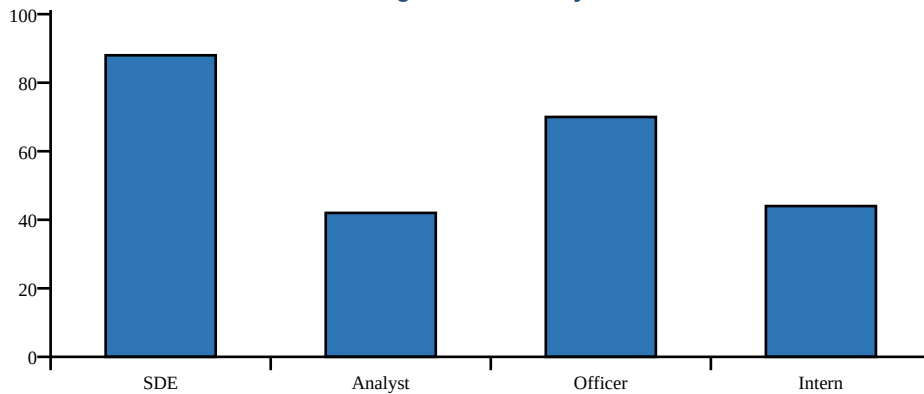
■ **Tip:** Microsoft values problem-solving approach over perfect solutions. Explain your thought process aloud. OS and DBMS knowledge is tested more than at other MNCs.

Section 3: Hiring Distribution by Role

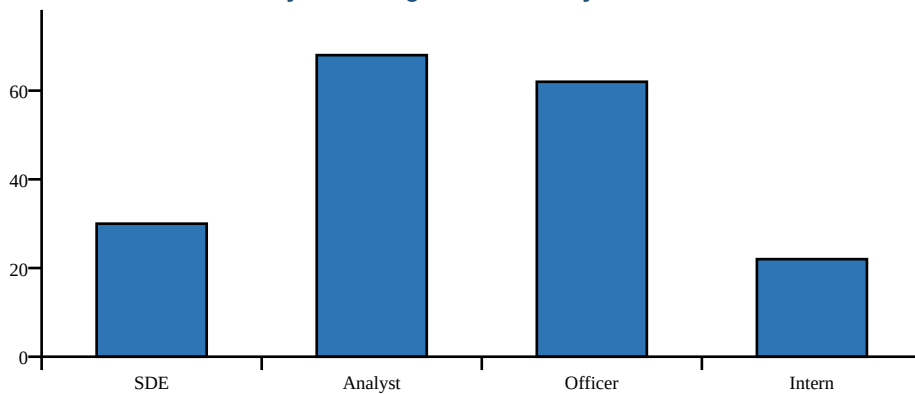
■ RAG Challenge — Image / Chart Content

Bar charts are IMAGE content — standard text extractors cannot read them. To answer queries like 'Which company hires the most Analysts?', students must either: (a) use a Vision-Language Model (VLM) to embed chart images, or (b) convert chart data to text/table before indexing. This section provides both the chart AND the underlying data for comparison.

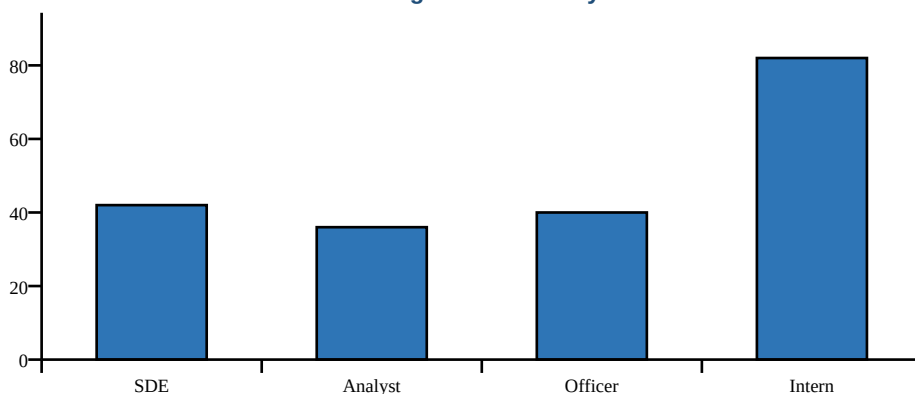
TCS — Hiring Distribution by Role

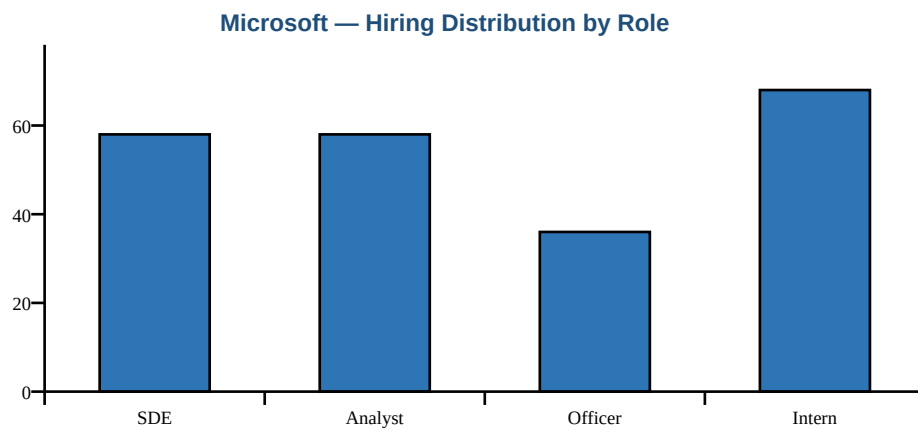
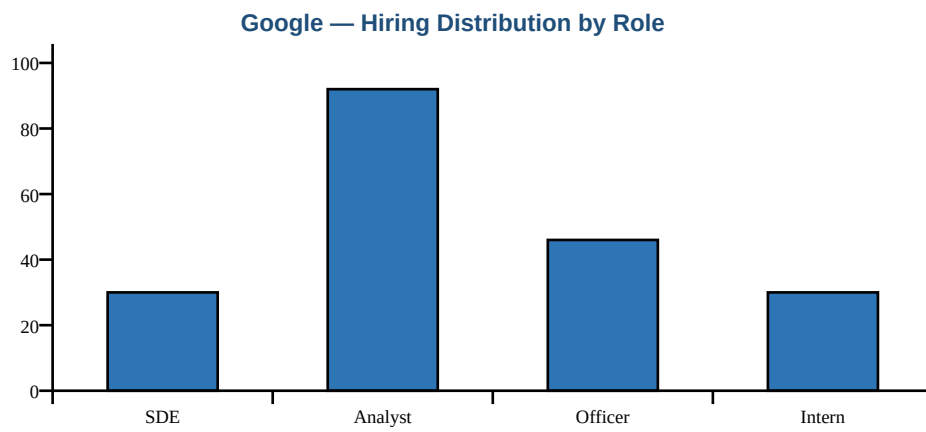


Infosys — Hiring Distribution by Role



Amazon — Hiring Distribution by Role





Hiring Distribution Data Table (Text representation of all charts above)

Company	SDE	Analyst	Officer	Intern	Total
TCS	88	42	70	44	244
Infosys	30	68	62	22	182
Deloitte	42	85	62	44	233
Accenture	25	22	52	68	167
Amazon	42	36	40	82	200
Flipkart	58	55	50	32	195
Google	30	92	46	30	198
Microsoft	58	58	36	68	220
Wipro	42	92	40	82	256
Cognizant	48	28	82	34	192
Capgemini	68	38	50	58	214
IBM	58	38	78	68	242
Adobe	42	80	62	48	232
Oracle	35	92	62	95	284
SAP	48	42	28	38	156
HCL	48	42	38	32	160
Tech Mahindra	58	28	58	30	174

Company	SDE	Analyst	Officer	Intern	Total
Qualcomm	25	38	82	78	223
Intel	48	48	42	48	186
Samsung R&D;	42	80	42	38	202

Section 4: Multi-Hop Reasoning Scenarios

■ RAG Challenge — Multi-document Synthesis

These queries CANNOT be answered from a single chunk. The RAG system must retrieve from multiple sections (eligibility table + hiring data + interview experience) and synthesise. This tests whether students implemented proper cross-document retrieval.

Q1: A student with CGPA 7.6 and 1 backlog wants the highest-paying job they qualify for.

Hop Type:	Hop-2: Eligibility + Package
Reasoning Chain:	Step 1: Filter companies where $CGPA \leq 7.6$ AND $backlogs_allowed \geq 1 \rightarrow \{\text{Deloitte, Amazon, Flipkart, Wipro, IBM, HCL, Tech Mahindra, Qualcomm, Samsung R\&D}\}$. Step 2: Sort by package $\rightarrow$ Amazon (28.6 LPA) is the highest. Answer: Amazon at 28.6 LPA.
Sources Required:	Eligibility Table + Package Column

Q2: Which company that uses Python in interviews offers the highest package?

Hop Type:	Hop-2: Tech Focus + Interview Prep
Reasoning Chain:	Step 1: Find companies with Python as tech focus $\rightarrow$ Google (42.0), Flipkart (25.3), Oracle (17.3), Intel (41.4). Step 2: Sort by package $\rightarrow$ Google (42.0 LPA). Answer: Google at 42.0 LPA.
Sources Required:	Tech Focus Column + Package Column

Q3: A student with CGPA 8.0 and 0 backlogs wants a company that hires many Analysts and pays above 20 LPA.

Hop Type:	Hop-3: Eligibility + Hiring + Package
Reasoning Chain:	Step 1: Filter $CGPA \geq 8.0$, $backlogs = 0 \rightarrow \{\text{Infosys, Accenture, Cognizant, Capgemini, SAP, HCL}\}$. Step 2: Cross-reference Analyst hiring $> 40 \rightarrow \{\text{Infosys(68), Cognizant(28-low), Capgemini(38)}\}$. Step 3: Filter package > 20 LPA $\rightarrow$ Infosys (42.9 LPA, 68 Analyst hires). Answer: Infosys.
Sources Required:	Eligibility Table + Hiring Table + Package Column

Q4: Which zero-bond companies offer more than 40 LPA?

Hop Type:	Hop-2: Bond + Package Trade-off
Reasoning Chain:	Step 1: Filter Bond = 0 $\rightarrow \{\text{TCS, Infosys, Microsoft, IBM, Intel}\}$. Step 2: Filter Package > 40 LPA $\rightarrow$ Infosys (42.9), Intel (41.4). Answer: Infosys (42.9 LPA) and Intel (41.4 LPA).
Sources Required:	Bond Column + Package Column

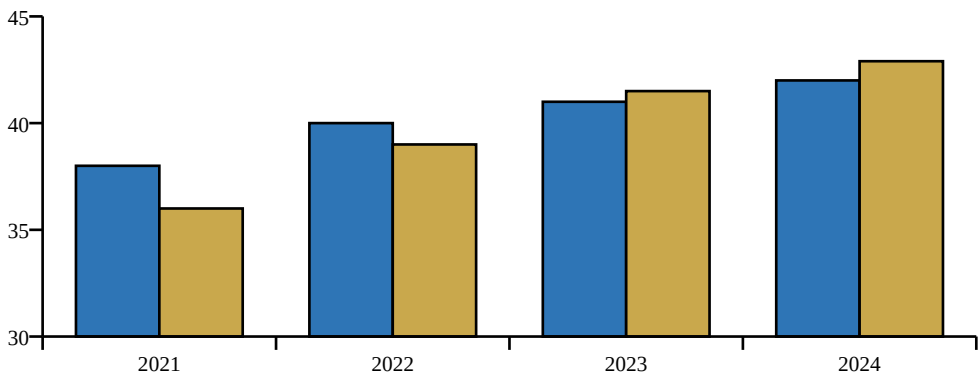
Section 5: Placement Trend Data (2021–2024)

■ RAG Challenge — Temporal Reasoning

Queries involving 'increase', 'decrease', 'best year', or 'trend' require reasoning across time-indexed data. Chunks must retain the year metadata to answer correctly. Without proper metadata tagging, time-based retrieval fails completely.

Company	2021 (LPA)	2022 (LPA)	2023 (LPA)	2024 (LPA)	3-Year Trend
TCS	3.6	3.8	4.0	4.1	↑ Steady growth
Infosys	36.0	39.0	41.5	42.9	↑ Strong growth
Amazon	22.0	25.0	27.0	28.6	↑ Consistent rise
Google	38.0	40.0	41.0	42.0	↑ Marginal growth
Deloitte	7.0	8.2	9.0	9.6	↑ Steady growth
Microsoft	19.0	20.0	21.0	21.4	↑ Slow growth
Wipro	24.0	25.0	25.8	26.1	↑ Slow growth
Cognizant	38.0	40.0	41.5	42.3	↑ Strong growth
Accenture	14.0	15.0	16.5	17.3	↑ Moderate growth
Flipkart	22.0	23.0	24.5	25.3	↑ Moderate growth

Package Trend: Google vs Infosys (LPA)



Sample temporal query: Which company showed the largest absolute package increase from 2021 to 2024? (Answer: Infosys — grew by 6.9 LPA, from 36.0 to 42.9)

Section 6: Conflicting Data — Hallucination Detection Test

■ RAG Challenge — Conflicting Information

This section intentionally contains CONFLICTING records for the same companies. A RAG system that simply retrieves the nearest chunk may return the wrong value without detecting the conflict. Students must implement conflict detection, or the LLM must be prompted to flag contradictions.

The following records were obtained from a DIFFERENT source (unofficial placement portal scrape). Values may differ from the official data in Section 1.

Company	CGPA (Official)	CGPA (Portal)	Package Official	Package Portal	Conflict?
TCS	7.5	7.0	4.1 LPA	4.5 LPA	✓ Both
Amazon	6.4	7.0	28.6 LPA	32.0 LPA	✓ Both
Google	7.4	7.5	42.0 LPA	45.0 LPA	✓ Both
Infosys	8.0	7.5	42.9 LPA	42.9 LPA	CGPA only
Microsoft	6.1	7.0	21.4 LPA	25.0 LPA	✓ Both

Expected RAG behaviour: When asked 'What is Amazon's CGPA cutoff?', the system should retrieve BOTH records and respond: 'There are conflicting records. The official criteria states 6.4, while the placement portal lists 7.0. Please verify with the official placement cell.' — NOT silently return one value.

Section 7: Overall Placement Statistics

Company	Avg Package	Max Offers	Min Offers	Avg CGPA Cutoff	Bond-free?
TCS	27.3	56	12	7.5	Yes
Infosys	27.2	149	30	8.0	Yes
Deloitte	8.4	127	22	7.7	No
Accenture	9.2	110	18	8.2	No
Amazon	20.5	87	15	6.4	No
Flipkart	12.7	113	20	7.8	No
Google	25.7	136	28	7.4	No
Microsoft	26.9	157	32	6.1	Yes
Wipro	22.2	111	19	6.7	No
Cognizant	34.2	70	10	8.4	No
Capgemini	29.1	95	14	7.1	No
IBM	18.4	88	16	7.5	Yes
Adobe	15.2	72	11	7.5	No
Oracle	13.8	102	18	7.7	No
SAP	17.5	60	9	8.4	No
HCL	21.3	80	13	8.4	No
Tech Mahindra	28.7	90	17	8.1	No
Qualcomm	33.2	78	12	7.2	No
Intel	32.5	68	10	7.0	Yes
Samsung R&D;	6.4	82	14	6.3	No

Section 8: Adversarial & Out-of-Corpus Test Queries

■ RAG Challenge — Graceful Fallback

These queries CANNOT be answered from this document. A well-built RAG system must respond with 'I don't have enough information in the provided documents' rather than hallucinating an answer. Teams should test their system against all queries below and log the responses.

#	Query	Type	Expected RAG Response
1	What is the exact date TCS will visit SVECW campus next?	Out-of-corpus	Fallback — no date information in corpus
2	How many students from SVECW got placed in Google last year?	Out-of-corpus	Fallback — no institution-specific placement count
3	What is the current Infosys stock price?	Real-time / Out-of-corpus	Fallback — no financial market data in corpus
4	Which company is better — Google or Microsoft?	Opinion / Subjective	Clarify it depends on student's profile; present objective comparison
5	Does Amazon offer work-from-home?	Out-of-corpus	Fallback — no work mode information in corpus
6	What programming language should I learn for TCS?	Partial — corpus has 'System Design' focus	System Design is tested; specific language not mentioned
7	Which company pays the most in the world?	Out-of-scope	Fallback — corpus covers only listed companies
8	I have a CGPA of 5.0. Which company can I apply to?	Edge case — below all thresholds	No company in this dataset has a CGPA cutoff ≤ 5.0

Section 9: Official Evaluation Query Set (30 Questions)

Judges will use these 30 queries to evaluate each team's RAG system during the hackathon. Queries are marked by difficulty and the RAG challenge they specifically test.

ID	Query	Difficulty	RAG Skill Tested
E1	What is the CGPA requirement for TCS?	Easy	Direct table lookup
E2	How many backlogs does Deloitte allow?	Easy	Direct table lookup
E3	What is the bond period for Amazon?	Easy	Direct table lookup
E4	Which technology does Flipkart focus on in interviews?	Easy	Direct table lookup
E5	What is the package offered by Google?	Easy	Direct table lookup
E6	Does Microsoft allow backlogs?	Easy	Boolean table query
E7	What rounds does TCS conduct?	Easy	Text retrieval
E8	Which programming language is tested at Amazon?	Easy	Text retrieval
M1	List all companies that allow at least 2 backlogs.	Medium	Multi-row filter
M2	Which companies require a CGPA above 8.0?	Medium	Threshold filter
M3	Which company has the highest package among IT service firms?	Medium	Category + sort
M4	Which companies are bond-free?	Medium	Boolean filter
M5	Compare TCS and Infosys on all eligibility criteria.	Medium	Multi-attribute comparison
M6	How many SDE roles does Amazon hire versus Google?	Medium	Chart/table comparison
M7	Which company hires the most Interns?	Medium	Hiring table aggregation
M8	What topics should I prepare for a Microsoft interview?	Medium	Text retrieval + synthesis
M9	Which company's package grew the most from 2021 to 2024?	Medium	Temporal reasoning
M10	Which companies use Python as the technical focus?	Medium	Column filter
H1	A student with CGPA 7.0, 1 backlog wants maximum pay with no bond.	Hard	3-condition filter
H2	Which Python-focused company hires the most Interns?	Hard	Join: tech + hiring
H3	For CGPA 8.0+, zero backlog students, rank companies by package.	Hard	Filter + sort
H4	Which company had conflicting CGPA data across sources?	Hard	Conflict detection
H5	Is the Amazon CGPA cutoff 6.4 or 7.0? Explain.	Hard	Conflict resolution
H6	Which company offers the best package-to-CGPA ratio?	Hard	Computed aggregation
H7	Compare Google and Amazon on all dimensions: eligibility, package, bond, trend.	Hard	Full synthesis
X1	What is TCS's campus visit date at SVECW?	Expert	Out-of-corpus fallback
X2	Should I join Google or Microsoft? Which is better for my career?	Expert	Opinion / subjective
X3	I have CGPA 5.0. Where can I apply?	Expert	Below-threshold edge case
X4	What is Infosys's current stock price?	Expert	Real-time out-of-corpus
X5	Which company in this dataset pays the highest in the world?	Expert	Scope boundary handling

Section 10: Recommended Chunking Strategy for This Dataset

This section guides students on HOW to chunk this specific document for optimal RAG performance. Different content types require different chunking strategies.

Content Type	Recommended Strategy	Chunk Size	Metadata to Store	Why
Company profile tables	Row-per-company serialisation	1 row = 1 chunk	company, section='eligibility'	Enables precise single-company retrieval
Interview experiences	Semantic / paragraph split	200–300 tokens	company, round_number, section='interview'	Deduplication needed first
Hiring charts (images)	Vision model caption → text	1 chart = 1 chunk	company, chart_type='hiring'	Images must be converted to text
Overall stats table	Full table as single chunk	Whole table	section='statistics'	Cross-company queries need full context
Temporal trend data	Row-per-company per year	1 row = 1 chunk	company, year, section='trend'	Year metadata enables temporal queries
Conflicting records	Keep both + add conflict flag	1 record = 1 chunk	source='official'/'portal', conflict=True	LLM must know both exist
Adversarial queries	Do NOT embed — use as eval	N/A	N/A	These are test questions, not corpus

Key Insight for Judges: Ask students to show their chunk count. An unoptimised pipeline on this document produces ~400+ chunks (due to 3× duplication). A well-optimised pipeline with deduplication should produce ~80–120 meaningful chunks.